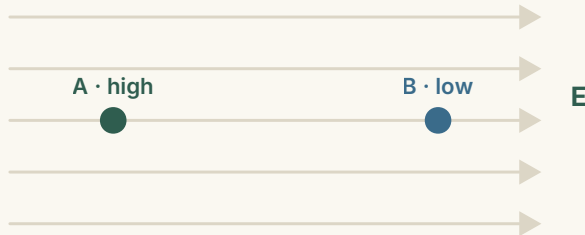


Voltage & Equipotentials

The “electrical height” between two points — and the surfaces that share it.

ESSENTIAL QUESTION

Moving from one point to another, why does each unit of charge gain the same energy — no matter which charge we carry?



● LESSON MAP

From a familiar hill to a map of the field.

01

ANALOGY

Potential plays the role of height in a field.

02

DEFINE

Voltage is the difference of potential between two points.

03

WORK

The work on a charge is $W = qU$.

04

SURFACES

Equal-potential surfaces map the whole field.

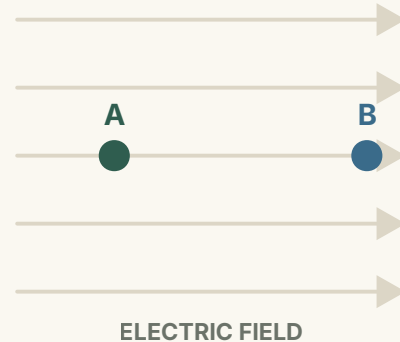
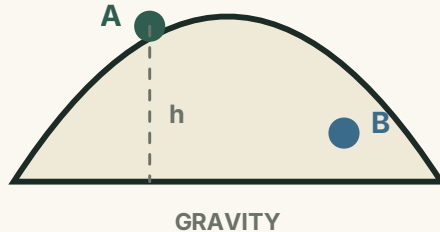
KEEP THIS IN MIND

Voltage is set by the **field** and the two points — not by the charge we test it with.

RECALL

Height is to gravity what potential is to a field.

You already know how a hill stores energy. The electric field works the same way.



DEFINE

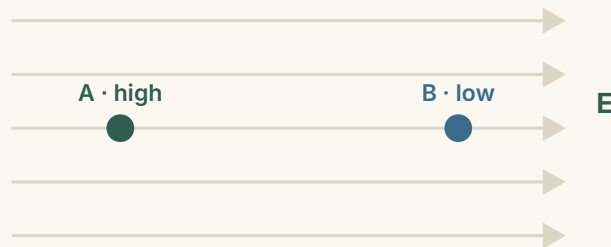
Voltage is the difference of potential.

Pick two points A and B. Subtract their potentials — that difference is the voltage.

FROM A TO B

$$U_{AB} = V_A - V_B$$

V_A and V_B are the energies per unit charge at each point; U_{AB} is measured in volts (V).



MODEL

Voltage is a signed scalar — read the sign.

Signed

$U_{AB} > 0$ means A sits higher than B; < 0 means lower.

No direction

A scalar, not a vector — the sign shows relative height.

Reversible

Swap the order and the sign flips: $U_{BA} = -U_{AB}$.

Reference-free

Shift every potential together and the difference is unchanged.

ALWAYS Name both points: U_{AB} and U_{BA} are different voltages.

DERIVE

The work on a charge is its charge times the voltage.

Start from energy, swap in the potential, and the test charge factors out.

$$W = qU_{AB}$$

WORKS IN ANY FIELD

No “uniform field” assumption was used — $W = qU$ holds for every electrostatic field, uniform or not.

CHECK

Voltage is fixed by the field, not your test charge.

A common trap: “since $W = qU$, doesn't a bigger charge mean a bigger voltage?”

1 Test with q

Between A and B the field does work $W = qU$.

2 Test with $2q$

Double the charge $\rightarrow$ double the work, $2qU$.

3 The ratio is the same

Voltage is decided by the source charges and the two positions — full stop.

THE POINT

W scales with the charge; U does not. Voltage belongs to the field.

APPLY

Four ways to find the work done on a charge.

Pick the one that fits what the problem gives you.

Definition

Force along the path: $W = Fd$. Uniform field only.

Energy change

From stored energy: $W = E_{pA} - E_{pB}$.

Voltage

From the potential drop: $W = qU$.

Work–energy

With other forces, via the work–energy theorem.

SCOPE Methods 2–4 work in any field; only the first needs a uniform field.

MEASURE

A handy energy unit falls out of $W = qU$.

Push one electron through 1 volt and see how much energy it gains.

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

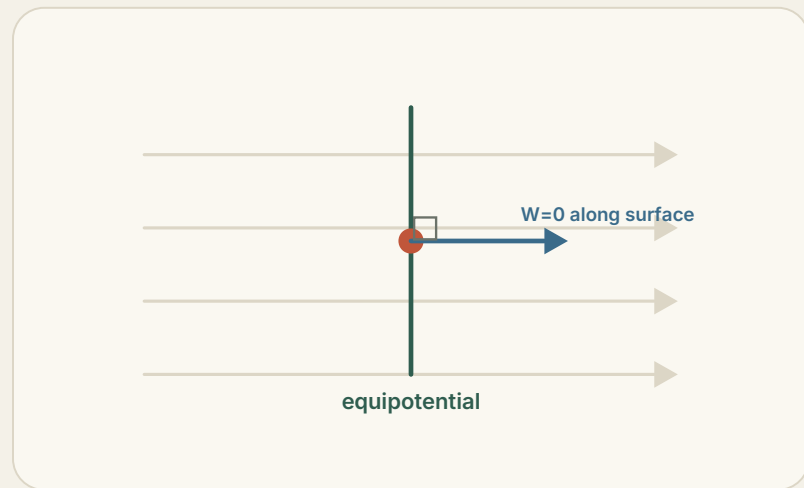
THE ELECTRON-VOLT

The energy an electron gains across a 1 V difference — the natural currency for tiny particles.

MODEL

Join points at one potential — an equipotential surface.

Like a contour line on a map: everywhere on it sits at the same electrical height.



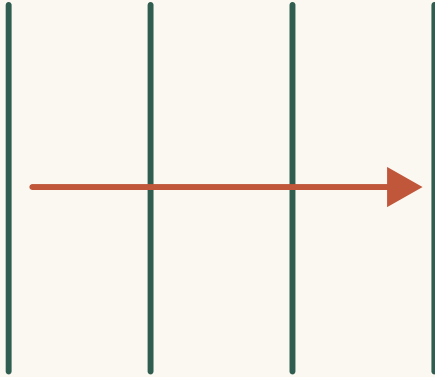
- 1 Perpendicular**
Equipotentials always cross field lines at 90° .
- 2 No work along**
 $W = q\Delta V = 0$, since the potential doesn't change.
- 3 Field points downhill**
The field runs from higher potential toward lower.

COMPARE

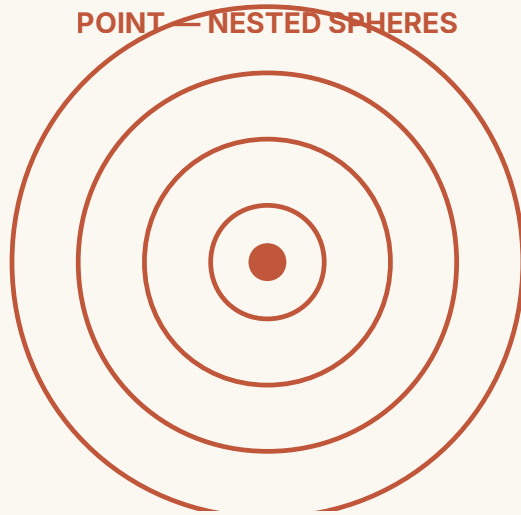
Two fields, two families of equipotentials.

The shape of the surfaces tells you the shape of the field.

UNIFORM — FLAT PLANES



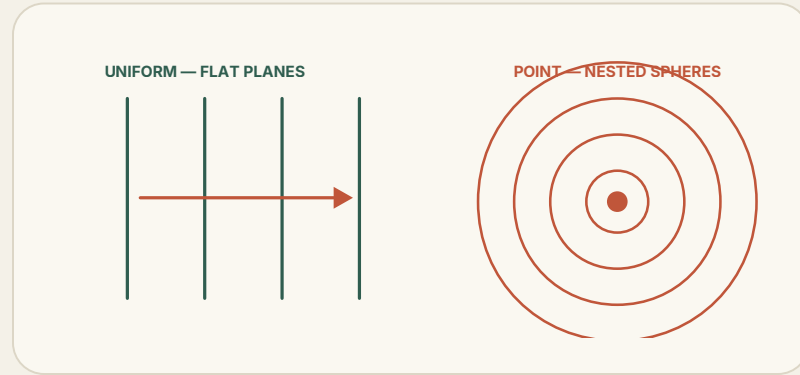
POINT — NESTED SPHERES



OBSERVE

Crowded equipotentials mean a strong field.

Draw the surfaces at equal voltage steps; where they bunch up, the field is strongest.



Equal voltage steps with smaller gaps $\rightarrow$ a larger field E .

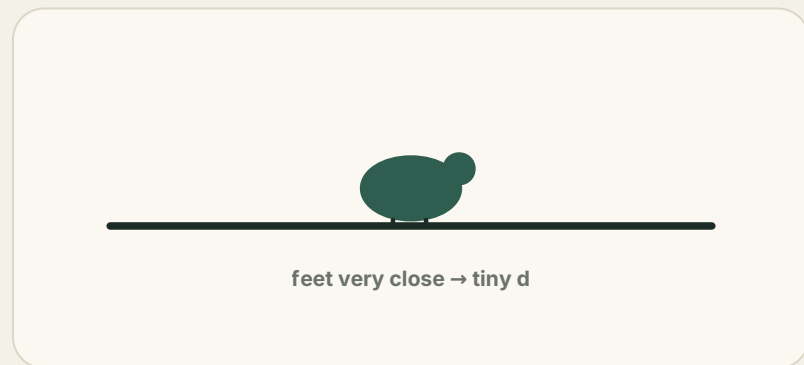
READ THE MAP

Tight surfaces = strong field; widely spaced = weak field.

APPLY

Why the bird on a high-voltage line is safe.

The wire may be at hundreds of thousands of volts — yet the bird is unharmed.



Both feet touch nearly the **same point**, so the voltage between them is tiny.

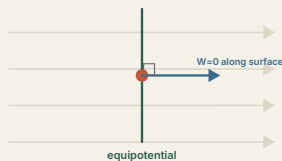
With a small U , the work qU on charges in the bird is tiny too — no harmful current flows.

● SUMMARY

Voltage is the field's **"height drop"** between two points.

IT SETS THE WORK ON ANY CHARGE

$$W = qU_{AB}$$



$U_{AB} = V_A - V_B$, fixed by the field — not by the charge you test with.

Surfaces of equal potential map the whole field at a glance.